

Intro to Inverse Kinematics

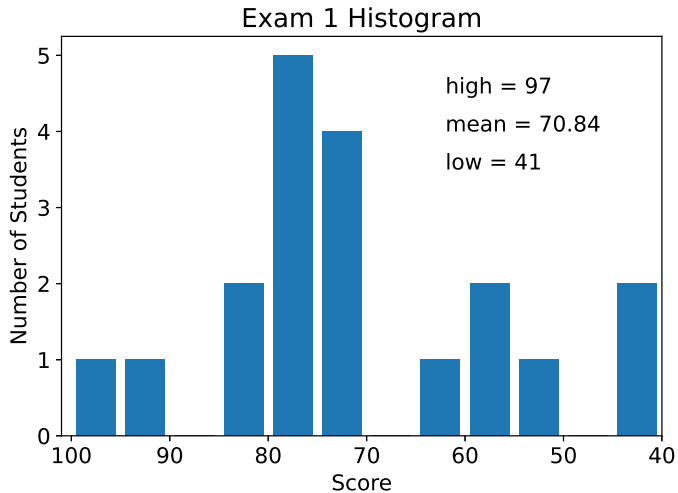
EGR 445/545

GVSU

Plan for Today

- exam 1 solution
- intro to inverse kinematics
- project time

Exam 1 Histogram



Intro to Inverse Kinematics

What do we mean by inverse kinematics?

- what is the definition of kinematics?
- what do we mean by inverse in this case?

Inverse Kinematics is Hard

- why?
- why can't I just make it easy?
- what did we have to do to find θ_1 and θ_2 for project 1?
 - was there an easier way?

Complaint of a former student

- “Stop talking so much about what makes a valid problem statement and give me a repeatable way to solve inverse kinematics problems.”
 - what are they really asking for?
 - why can't I meet this request?
 - why do *you* need to know what makes a valid inverse kinematics problem statement?

You can't solve an unsolvable problem

- your customer *probably* doesn't have better technical skills than you
- your boss *probably* doesn't have better technical skills than you
- if someone asks you to do something that cannot be done, what are you going to do?

Possible causes of unsolvable problems

- reachable workspace
- dextrous workspace
- # DOFs vs. # of joints

An Inverse Kinematics Process: The Best I Can Do

- can you locate the wrist?
 - do you have enough information?
- what can you learn from a top view?
- what can you learn from a side view?
 - what frame is the side view in?

Industrial Robot Inverse Kinematics

- how many joints do most industrial robots have?
 - why?
- what would a valid problem statement for inverse kinematics need to include?
 - how would we solve such a problem?

What makes a valid inverse kinematics solution?

- find all solutions
- do not use numeric search algorithms
 - why not?
- find explicit solutions
- only trust `atan2`

Geometric vs. Algebraic Solution

- but I'm awesome at algebra. . .
 - can you find an algebraic solution for project 1?
- see section 4.4

Intro Example

- consider a simplified project 2 robot